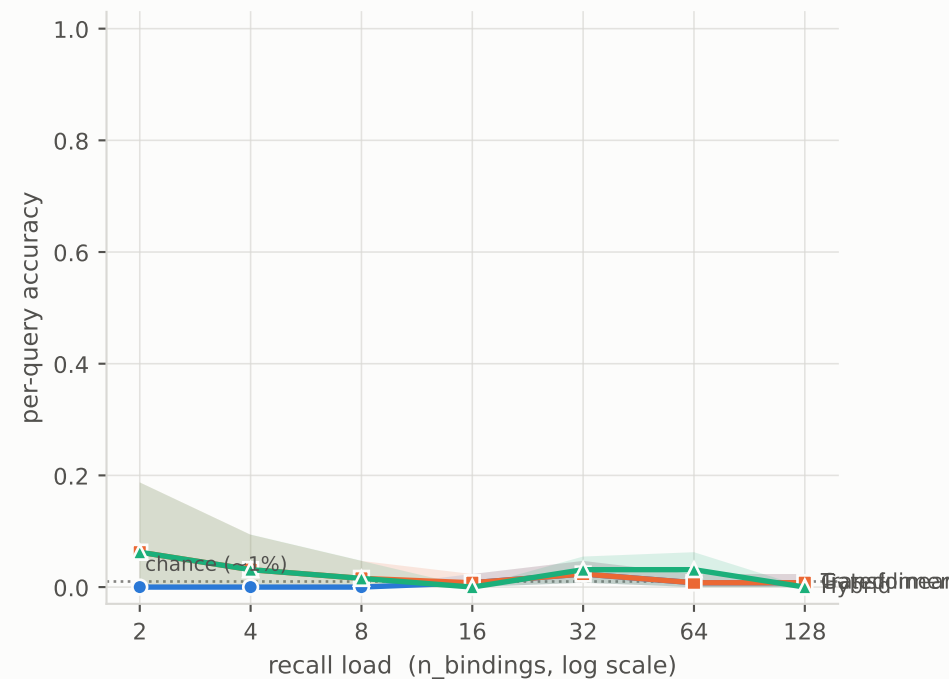
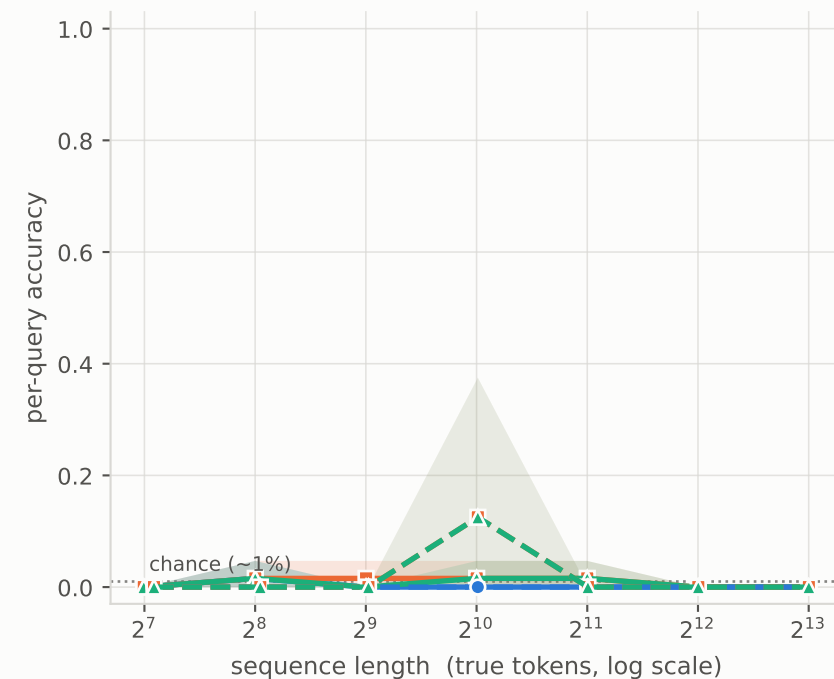


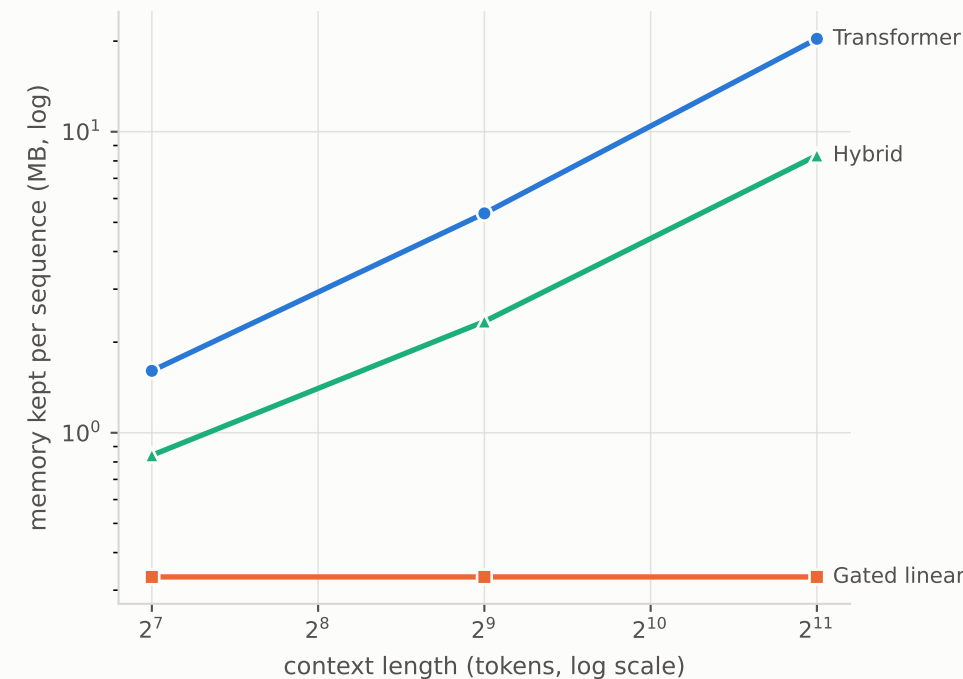
A • Capacity
multi-query recall, sequence length held in-distribution



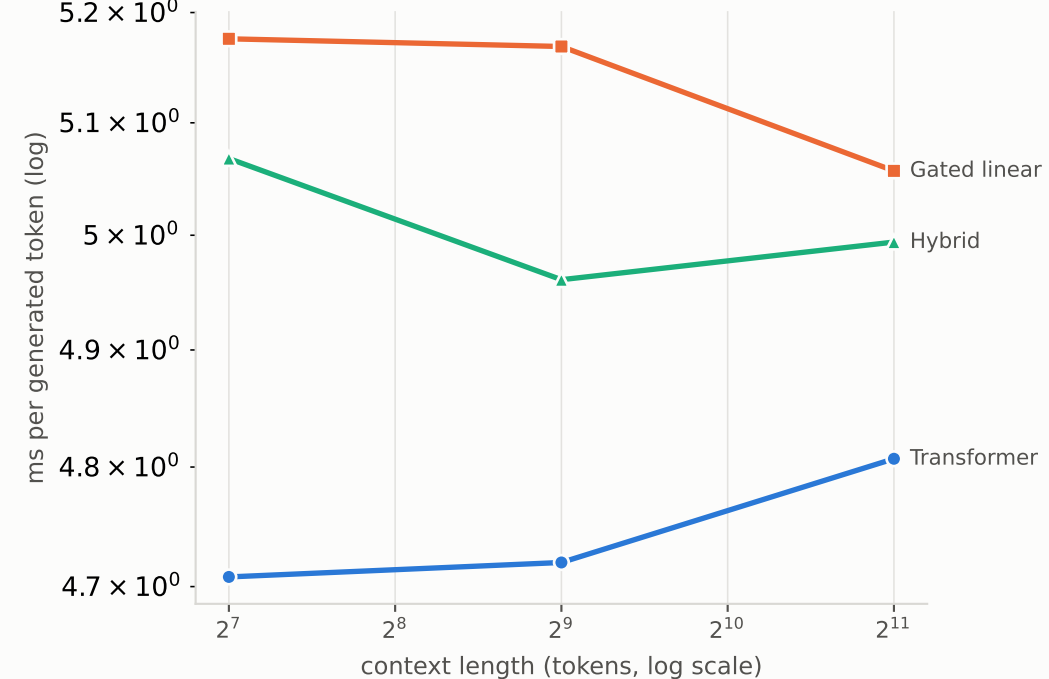
B • Length generalisation
solid: associative recall • dashed: state-track control



C • Decode memory
KV cache vs fixed state, batch 1



D • Decode latency
one token at a time, batch 1



Transformer (dense) Gated linear (fixed state) Hybrid